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Atomic input for modeling ionic mixtures in astrophysical plasma

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Abstract Accurate atomic data are known as a key ingredient for modeling the physical behavior of a wide range of plasma throughout the universe. When these data are combined with simulations of ionic mixtures, they (may) help infer the temperature, electron density, and ionization state of the plasma under non-ideal plasma conditions. We here report about an expansion of JAC, the Jena Atomic Calculator, in order to encourage the modeling of ionic mixtures and their thermodynamic properties under different plasma conditions. Whereas emphasis has first been placed upon simulations of ionic compositions in (ideal) local thermodynamic equilibrium, various features are prepared in this code to account for non-ideal conditions, such as plasma screening contributions, the ionization potential depression (IPD) or some non-classical statistics. To demonstrate these features of JAC, we have calculated the mean charge and charge-state distribution of a hot carbon plasma as well as some ionic mixture from the sun's photosphere at selected temperatures T and mass densities ρ_m. These computations show a significant shift in the mean charge distribution, if the IPD is incorporated by means of a Stewart–Pyatt plasma model. Since the JAC toolbox is based on Dirac's relativistic equation, the present expansion of the code will facilitate also the treatment of ionic mixtures with medium and heavy elements as relevant for neutron-star mergers and elsewhere.

1 Introduction

Astrophysical plasma occurs under very different conditions throughout the universe, including stars, planetary nebulae, interstellar and intergalactic media, and up to the observed remnants of supernovae. In particular, supernovae and the stellar nucleosynthesis have often been mentioned in the literature [1–4]. More recently, the production of these heavy elements through neutron-star mergers and extended r-process chains have attracted a lot of interest [5,6] and is in the focus of this volume [7].

To model such plasma over a good range of temperatures and matter densities, a large amount of "atomic input" is typically needed. Indeed, this input is quite fundamental in order to interpret observations and to comprehend the behavior of plasma within diverse environments. Besides level energies and ionization potentials, plasma rate coefficients (of various kind) are required to model how ions, atoms or molecules gain or loose electrons under different temperature and density conditions, cf. Fig. 1. The (detailed) interplay of all this input eventually determines altogether the ionization balance of elements in stars, nebulae, and accretion disks [8,9].

While, on a microscopic scale, plasma just consists of free electrons and ions, one typically wishes to understand how the (nuclear as well as the) electronic structure and processes affect the behavior of astrophysical objects at their macroscopic scale. A link between these microscopic and macroscopic viewpoints is often sought by modeling ionic mixtures in a – more or less – ideal local thermodynamic equilibrium (LTE). In these mixtures, all singly and/or multiply ionized atoms usually share the same temperature (and matter distribution) over certain regions of space and can thus be modeled by (a set of) common thermodynamic properties, such as the pressure, internal energy density or several others.

In practice, however, these ionic mixtures can usually not be handled as an ideal gas but require to account for the Coulomb interaction among the particles [10], the screening of electrons due to modified ionization potentials [11,12] or to even account for (sophisticated) correlation functions [13,14]. These corrections become relevant especially if the de Broglie wavelength of the plasma particles becomes comparable with their relative distance. Formally, these interactions and screening contributions are commonly captured into the equation of state (EOS) as the central relation-

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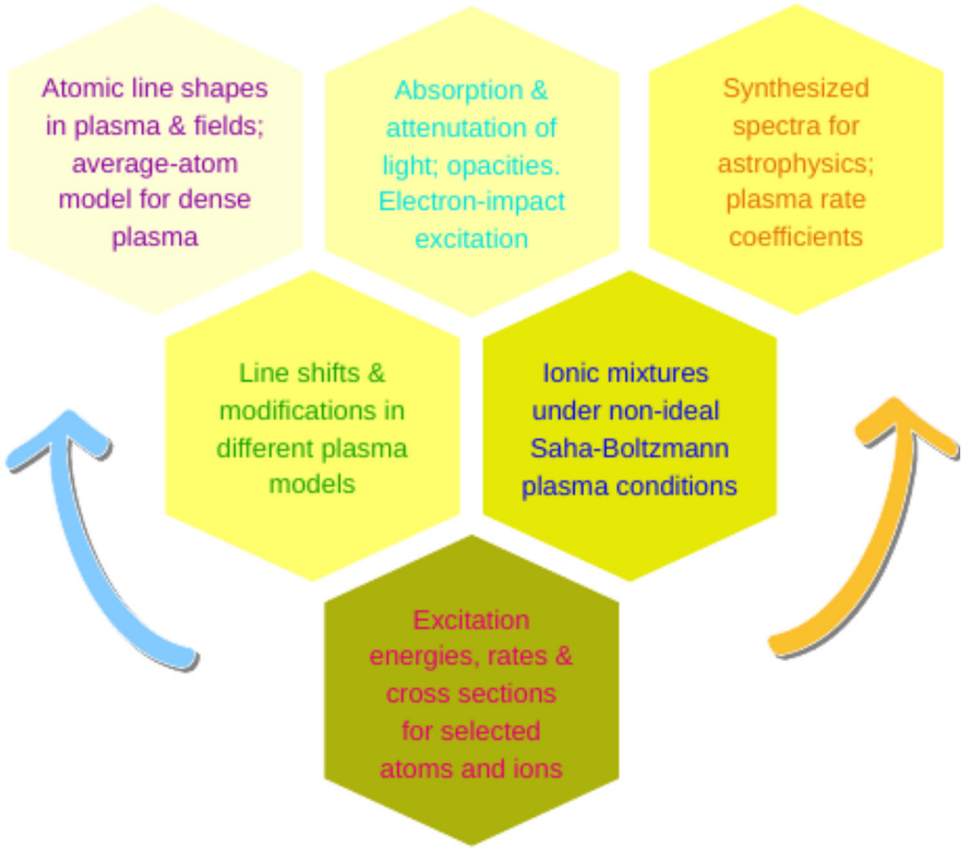

Fig. 1 Needs for plasma modeling based on *(ab-initio)* atomic input. The concept (and distinction) of different plasma scenarios and computations enables one to link the electronic-structure calculations of individual atoms and ions to their use and impact in astro and plasma physics. Moreover, various kinds of plasma computations not only require "atomic input" but help serve other communities in physics, science and technology

ship between pressure, volume, temperature and the internal energy of ionic mixtures. Finding this EOS has remained a challenge for computational plasma physics but need to be considered under quite different temperature and matter-density conditions.

Several codes exist in the literature to model the thermodynamic properties of ionic mixtures [13,15–17]. They are based on different assumptions for – generating and truncating – the level structure of ions, the choice of interaction potentials, the use of additional empirical data or how certain collective plasma contributions are taken into account. The diversity of models and their *ad-hoc* combination, which are considered to explain selected features in the behavior of plasma, is hereby enormous. Central to most of the codes is the quantum-mechanical treatment of the atomic level and, hence, line structure in different plasma. The electronic level structure of atoms and ions stands also in the focus of Jac, the Jena Atomic Calculator [18,19], that help compute level energies, energy shifts as well as various cross sections, rates and rate coefficients. In this work, we have expanded this toolbox in order to assist plasma simulations which are based on electronic-structure calculations of the underlying atoms and ions. In particular, we here aim to formalize (and simplify) the computation of Saha-Boltzmann (SB) equilibria of ionic mixtures under different plasma conditions, and by supporting mixtures of heavy elements. When compared to other codes, the Jac toolbox facilitates the handling as well as the incorporation of additional corrections for non-ideal plasma. This expansion makes Jac an excellent tool to analyze and estimate various thermodynamic plasma properties under different plasma conditions.

In the next section, we shall first recall the concept of SB solvers for ionic mixtures. Apart from their characterization and use under LTE condition, this outline includes a brief account on their thermodynamic properties and the ionization-potential depression (IPD) in dense plasma. The implementation of such ionic mixtures into Jac is made via (the data structure of) a `Plasma.Computation`. Section 3 gives a short overview of the Jac toolbox, along with a clear distinction of different plasma scenarios, data types as well as the use of the newly implemented SB solver. In Sect. 4, we then apply this solver to simple ionic mixtures to demonstrate how readily the charge distribution of the ionic species or their mean charge state can be extracted for different plasma conditions. Results are shown for rather a hot carbon plasma as well as some ionic mixture from the sun's photosphere at selected temperature T and mass density ρ_m. Finally, a few conclusions and a short outlook are given in Sect. 5.

$$n_{l_o}^{(\mathrm{eq})}(T;\ Z,\ A,\ q_o) = n(T;\ Z, A) \frac{g_{l_o}(Z,\ A,\ q_o)\ \exp\left[-\beta\left(\mu_e(T;\ n_e)\, q_o\ +\ E_{l_o}(Z,\ A,\ q_o)\right)\right]}{\sum_{q,l} g_l(Z,\ A,\ q)\ \exp\left[-\beta\left(\mu_e(T;\ n_e)\, q\ +\ E_l\,(Z,\ A,\ q)\right)\right]} \tag{1}$$

$$n_e^{(\mathrm{eq})}(T) = \sum_{\{[I(Z,A)]\},\, q,\, l} n_l\,(T;\ Z,\ A,\ q)^{(\mathrm{eq})} \cdot q \tag{2}$$

2 Implementation of ionic mixtures in non-ideal Saha-Boltzmann (SB) equilibrium

2.1 Ionic mixtures in local thermodynamic equilibrium

In the astrophysical context, plasma is often described in terms of some "chemical picture", in which ionic mixtures with given isotopic abundances interact with each other. In this picture, the goal of plasma simulations is to understand how these ions, atoms and molecules coexist under certain (common) thermodynamic conditions, such as temperature, pressure or the overall matter density, and which electronic states of the atoms and ions are populated in a plasma. More formally, such an (ionic) mixture refers to a set of ions $\{I^{q+} \equiv [I(Z,\ A)]^{q+}\}$ and N_e free electrons, where each ion is characterized by its nuclear charge Z, an atomic mass number A as well as the charge state $q+$. This notation implies that all ions are associated with a pre-selected set of isotopes $\{(Z,\ A)\}$ which contribute with some fraction $0\ \leq\ x\ \leq\ 1$ to the overall mixture; it also enables one to deal either separately with (individual) isotopes of an element or to just keep different isotopes together by means of a *mean* mass number A.

Under equilibrium conditions, we can use the number densities n [i.e. the particles of certain kind or population/cm^3] in order to model a mixture in terms of the densities $n(Z,\ A)$ of the isotopes (isotopic fractions) and n_e, the associated electron number density. If the plasma is assumed to be neutral, these number densities fulfill the simple relation $\sum_{\{I(Z,A)\}} n(Z,\ A)\ -\ n_e\ =\ 0$. For each isotope, moreover, different (classes of) charge states $q+$ need to be distinguished, each with some ground-state energy E_1 and an ordered list $(l\ =\ 1,\ 2,\ ...)$ of ionic levels that exhibit a (unique) energy E_l, degeneracy g_l as well as, for a given plasma model and conditions, an (ionic) level number density $n_l\ \equiv\ n_l\,(T;\ Z,\ A,\ q)$.

Different chemical pictures have been utilized by the community to deal with ionic mixtures in different astrophysical environments. Whereas the (well-known) SB picture is effective for weakly-coupled plasmas in LTE, strongly-coupled plasma models are required, if the matter density or pressure increases like in white dwarfs. Under these latter conditions, the chemical picture from above can still be useful but must account for the strongly non-ideal character of the plasma owing to its high density and the strong Coulomb interaction among the ions and electrons. This non-ideal character of the plasma can be treated, for instance, by employing a Debye shielding to model the "cloud" of oppositely charged surrounding particles or by making use of even more advanced models.

For a given ion number density n_i (per unit volume), a SB solver generally aims to determine the detailed ion and level composition by minimizing the total free energy of the plasma. Most readily, perhaps, this minimization can be formulated in terms of the chemical potential $\mu_l(T)\ \equiv\ \mu_l\,(T;\ Z,\ A,\ q)$ of the ionic levels at plasma temperature T

$$\mu_l(T;\ Z,\ A,\ q) = -\,k\,T\ \ln\left(\frac{g_l\,(Z,\ A,\ q)}{n_l\,(T;\ Z,\ A,\ q)\ \ [\Lambda(T;\ Z,\ A)]^3}\right) + E_l\,(Z,\ A,\ q)\,.$$

In this expression, $\Lambda(T;\ Z, A)$ refers to the thermal (de Broglie) wavelength of the associated isotopic fraction. From these chemical potentials, the charge-state distribution and mean charge state is then readily obtained, along with other thermodynamic properties.

We can make this task for obtaining the level populations at equilibrium explicit in terms of the two conditions

See Eqs. (1–2) above

that need to be fulfilled self-consistently (with $\beta\ =\ 1/k\,T$) for each ionic level number density $n_l\ =\ n_l\,(T;\ Z,\ A,\ q)$. These equations relate the level number densities to each other via the – temperature and matter-density dependent – partition function and, thus, accounts for the isotopic composition of the ionic mixture. At equilibrium, the mean charge state of all isotopic fractions $\{(Z,\ A)\}$ at temperature T is $\langle q(T;\ \{Z,\ A\})\rangle\ =\ n_e^{(\mathrm{eq})}\,(T)\,/\,n_i$. Not much need to be said about the use of such SB solvers which have been discussed frequently in the literature [15,20]. They are often applied to compute the ion population of different isotopes or for interpreting the (spectral) emission and absorption in plasma. A SB solver can also show how the thermal energy is distributed in the plasma and may, thus, provide insight into the stellar evolution and nucleosynthesis. Hereby, obviously, many of these applications require a simple control (and handling) of different ionic mixtures, the re-use of atomic data or the specification of different plasma conditions, to name just a few. With the present expansion of JAC, we aim to obey these needs and to advance these SB equilibrium studies towards the medium and heavy elements.

2.2 Thermodynamic properties of ionic mixtures

From the (mean) charge-state distribution of a given ionic mixture, several thermodynamic properties can be derived.

For example, the internal energy density of a SB mixture at temperature T is given by [21]

$$u(T) = \frac{3}{2} n_i \, k \, T \, (1 + \langle q(T;\, Z,\, A) \rangle) + \sum_{\{[I(Z,A)]\},\, q,\, l} n_l^{(\mathrm{eq})} (T;\, Z,\, A,\, q) \; E_l \, (Z,\, A,\, q).$$

A similar expression for the enthalpy $H = U + P\,V$ has been utilized in the literature [22] to understand the energy transfer in processes like heating or cooling in astrophysical systems, while the Gibbs free energy may help determine the spontaneity of reactions and phase transitions within ionic mixtures [23,24].

To model astrophysical plasma under quite different conditions, any (new) implementation of a SB solver should be flexible and computationally efficient with regard to different model extensions. Desirable corrections, which have been discussed at several places in the literature, include:

- IPD as a mean to approximate the contributions to the plasma free energy due to both, the interaction and polarization among the free electrons [20].
- a Fermi-Dirac statistics for dealing with the free electrons in dense plasma in addition to the classical Maxwell-Boltzmann statistics of ions and (neutral) particles; such a modified statistics affects the chemical potential and partition function of the free electrons [25].
- the account of selective radiative excitation and/or recombination processes that may *effectively* modify the charge-state distribution due to local external fields.

All these corrections arise from modifications of the EOS due to the non-ideal gas, i.e. from modified particle interactions at high densities and temperatures, or due to the occurrence of additional electric fields. Indeed, such an enhanced modeling of ionic mixtures will be crucial for understanding the behavior of plasmas in diverse environments, such as stellar atmospheres, accretion disks or interstellar media. Further work is certainly needed to explore how these and similar corrections affect the modeling. As an example, we next just summarize how the depression of the ionization potential can be treated in a SB solver.

2.3 Ionization potential depression. Truncation of ionic level data

In (non-ideal) plasma, the ionization potential of ions in charge state $q+$ is typically reduced owing to the Debye screening of the electron-nucleus interaction, if the electrons and ions come close to each other. This lowering of the potential, i.e. the reduced energy to ionize an atom or ion in a plasma relative to its ionization energy in the isolated system, is known as IPD or continuum lowering and "shifts" the weakly-bound electrons into the continuum without additional energy. In practice, however, such a *lowering* of the ionization potential generally refers to an increase of its total energy $E_{\mathrm{tot}}^{(q_o)} \rightarrow E_{\mathrm{tot}}^{(q_o)} - \sum_{q=q_o}^{Z-1} \Delta\, I_p(q \rightarrow q+1) > E_{\mathrm{tot}}^{(q_o)}$ of all levels of charge state $q+$, and which then directly affects the partition functions in the self-consistent computation of the SB equilibrium. Whereas such an IPD enables one to capture the plasma contributions simply into a set $\{\, \Delta I_p(q \rightarrow q{+}1) \,\}$ of uniform energies shifts, it assumes that the plasma has no other effect upon the electronic structure of the ion. Since the energy levels that are "moved" into the continuum of the next higher charge state refer to electrons, which rapidly de-localize, these levels are typically neglected in the partition function, improving its convergence.

In the implementation below, we account for the IPD of ionic mixtures by supporting a few selected plasma models. Among these models, the Debye-Hückel, ion-sphere or Stewart-Pyatt plasma models are perhaps best known to the community. In our implementation, these models first of all define a "frame" to be selected by the user, while their detailed parameters are determined later only in course of the self-consistent calculations. For a weakly-coupled plasma, the static Debye-Hückel model is usually applied and leads to the shift

$$\Delta I_p^{(\mathrm{DH})} (Z,\, A;\, q \rightarrow q+1) = -\frac{e^2}{4\pi\, \epsilon_o} \; \frac{(q\, +\, 1)^2}{\lambda_D}$$

with the Debye (screening) length

$$\lambda_D = \left[\frac{e^2}{4\pi\, \epsilon_o} \, (4\pi\, \beta) \left(n_e \, + \sum_{\{[I(Z,A)]\},\, q,\, l} n_l \, (T;\, Z,\, A,\, q) \cdot q^2 \right) \right]^{1/2}.$$

This screening length modifies the attractive nuclear interaction because of the neutrality of the overall plasma and tends to $D \rightarrow \infty$ for isolated atoms or ions, i.e. for $n_e \rightarrow 0$ and $\exp(r/D) \rightarrow 1$. The Debye-Hückel model has been commonly applied also in order to incorporate plasma effects into atomic structure calculations [26–28] but is often suitable only at rather low density and high temperature, and if the Debye length D is (much) larger than the size of the ions. Table 1 summarizes the basic assumptions and plasma conditions under which the different models are considered to be appropriate.

In Sect. 3 below, we shall explain how the IPD can be included in solving the SB equilibrium for ionic mixtures of different kind and complexity.

Table 1 Basic assumptions and conditions of the implemented mean-field plasma models in JAC

Model	Assumptions and conditions
Debye-Hückel (DH)	Weakly-coupled plasma; quasi-neutral; neglects all magnetic contributions; frequently applied in atomic structure calculations, though it remains under debate. Not appropriate for high-density plasma
Ion-sphere (IS)	Strongly-coupled plasma; Coulomb interaction dominates over thermal motion; high-density plasma with correlated structure; frequently applied in warm dense matter and astrophysical plasma
Stewart Pyatt (SP)	Wide range of plasma temperature and densities; attempts to interpolate between the DH and IS models

2.4 The Hummer-Mihala (HM) approach to solving ionic mixtures in astrophysics

The EOS of stellar envelopes describes the relationship between the (thermodynamic) key properties of the plasma, such as pressure, temperature, density and its ion composition. In stellar physics, the goal often is to determine these properties for the outer layers of the stars, just below of their atmosphere. For an ionic ensemble of different elements and charge states with, say, $T \lesssim 10^7$ K and $p \lesssim 10^2$ g cm^{-3}, Hummer and Mihalas [29] developed a statistical approach to the EOS of stellar envelopes, in which the free energy of the ionic ensemble is minimized in order to ensure thermodynamic consistency among the isotopic fractions. In this approach, all ions are treated as a non-ideal gas, whereas the individual particles in the plasma still retain their identity. Indeed, this approach is based on the chemical picture from above, in which the plasma particles interact pairwise with each other in terms of cross section and rates as assigned to different atomic processes. The HM approach has been found very useful and justified if the EOS need *not* to predict all the (bulk) thermodynamic properties of the gas but should just provide a reasonable distribution of the excited and ionized states of the atoms and ions from the gas.

In their later work, Mihalas and coworkers [30] applied the HM approach to evaluate the thermodynamic quantities for a few selected astrophysical mixtures at a rather elaborate temperature and density grid. With its careful treatment of non-ideal effects, the partition functions as well as its ionic mixtures, this approach has become a cornerstone in modeling stellar plasmas and for understanding the properties of rather different astrophysical objects. In particular, this HM approach provides a consistent framework for handling multi-component ionic mixtures, where multiple ionization states of elements coexist. The flexibility of the HM approach has made it applicable to both main-sequence stars as well as compact objects (e.g., white dwarfs).

3 Making plasma computations with JAC

3.1 A brief overview of the JAC toolbox

The JAC toolbox [18] has originally been developed for calculating atomic (many-electron) interaction amplitudes, properties as well as several excitation, ionization and capture processes for most open-shell atoms and ions [31,32]. This toolbox is based on the Dirac-Coulomb (-Breit) Hamiltonian and the multi-configuration Dirac-Hartree-Fock method [33,34] in order to represent the atomic states in terms of symmetry-adapted configuration-state functions (CSF). It also aims to calculate the rates and cross-sections of atomic processes within a single computational framework and, hence, to ensure consistency of the generated data. Since JAC is internally based on Dirac's equation, the theoretical framework is appropriate for modeling all elements across the periodic table, including the medium and heavy ones. This toolbox has been described elsewhere [35] and can readily be downloaded from the web [36].

The JAC code has been implemented in Julia, a new programming language for scientific computing. This language takes advantage of various (modern) programming features, such as dynamic types, multiple dispatch, just-in-time compilation of code, dynamic code loading or garbage collection [37,38], to recall just a few. It also comes with an expressive *type system* that allows optional type annotations in order to support an offensive code specialization against run-time types, if required. In Sect. 3.3 below, we shall discuss a few data types useful for plasma simulations.

To obey especially the needs from the atomic spectroscopy community [39–41], we have recently expanded JAC in order to support (so-called) `Cascade.Computation`'s of different kind [42], including the stepwise relaxation of inner-shell holes [43,44], the computation of dielectronic [45] and radiative (plasma) rate coefficients [46], or electron-impact processes [47]. These cascade computations can be comprised into a few well-distinguishable (atomic) cascade schemes, even if the underlying physics and computational treatment turns out to be different. However, all these schemes have in common that they are based on lists of explicitly specified electron configurations and that they typically request for generating a rathe large number of level energies and wave functions. A similar distinction of different (plasma) schemes will be made below to support plasma computations and to open this toolbox towards astro and plasma physics. The goal is to provide an interface which is equally accessible for researchers from astrophysical mod-

```
Plasma.Computation
  struct Plasma.Computation
    ... defines a type for defining different simple plasma computations for atoms and ions, based on
        a given set of reference configurations. It also enables the user to consider different atomic
        processes under plasma conditions, if the plasma is modeled empirically. The plasma environ-
        ment is typical described in terms of its temperature, matter density and by a few other
        parameters.

  + scheme          ::AbstractPlasmaScheme      ... Scheme (kind) of the plasma computation.
  + nuclearModel    ::Nuclear.Model             ... Model, charge and parameters of the nucleus.
  + grid            ::Radial.Grid               ... The radial grid to be used for the computations.
  + refConfigs      ::Array{Configuration,1}    ... A list of non-relativistic configurations.
  + asfSettings     ::AsfSettings               ... Provides the settings for the SCF process
                                                    (under plasma conditions) and for the CI.
  + settings        ::Plasma.Settings           ... General settings for the given computation.
```

```
  abstract type Plasma.AbstractPlasmaScheme
    ... defines an abstract type to distinguish different kinds of plasma computations; see also:

  + struct AverageAtomScheme      ... to perform an average-atom computation.
  + struct LineShiftScheme        ... to compute the energy shifts and properties of lines in some
                                      selected empirical plasma model.
  + struct SahaBoltzmannScheme    ... to compute the chemical potentials and level population of
                                      ionic mixtures in Saha-Boltzmann equilibrium.
```

```
struct Plasma.SahaBoltzmannScheme <: Plasma.AbstractPlasmaScheme
   ... a struct to compute the chemical potentials and level population of ionic mixtures for a
       Saha-Boltzmann equilibrium, along with their thermodynamic properties.

 + plasmaModel     ::Basics.AbstractPlasmaModel
     ... A plasma model that "constrains" the Saha-Boltzmann equilibrium densities by some
         additional conditions, for instance, due to IPD or others. Typically, the detailed
         parameters of these models are determined only in course of the computations.
 + calcLTE         ::Bool                  ... True, if the SB equilibrium is to be calculated.
 + printIonLevels  ::Bool                  ... True, for printing information about ionic levels.
 + qRange          ::UnitRange{Int64}      ... Range of charge states q to be taken into account.
 + maxNoIonLevels  ::Int64
     ... (maximum) number of ionic levels for any (individual) charge state of ions in the mixture
         to be taken into account into the partition function.
 + NoExcitations   ::Int64
     ... Number of single, double, ... excitations to be taken into account with regard to the
         reference (ground) configuration of the ions; usually, NoExcitations = 1,2 is applied to
         keep the computations feasible.
 + upperShellNo    ::Int64
     ... upper-most princicpal quantum number n for which orbitals are taken into account into the
         partition function of the ionic-level computations; often chosen as upperShellNo= 4..10.
         If ionic-level data are given already as input, this number help distinguish of whether
         these data are to be accepted internally.
 + isotopicMixture ::Array{Basics.IsotopicFraction,1}
     ... List of (non-normlized) isotopic fractions that form the requested ion mixture; the
         fractions will first be always renormalized to 1 in course of the Saha-Boltzmann LTE
         computations.
 + isotopeFilenames      ::Array{String,1}
     ... set of files names from which the ionic-level data can be read in for the different
         isotopes (Z,A) in the mixture.
```

Fig. 2 Selected abstract and concrete data structures in JAC that help compute the mean charge state (distribution) and thermodynamic properties of an ionic mixture under different plasma conditions. See text for further explanations

elling, observational astrophysics or code developers, and to provide and make use of atomic input of dissimilar kind.

3.2 Steps towards plasma computations with JAC

To approach this goal, let us distinguish and select different plasma scenarios that (may) each cover certain classes of plasma models and/or observations but leaves otherwise the electronic levels structure of the underlying atoms and ions (more or less) intact. All these scenarios usually come also with a set of characteristic assumptions, properties and measurements. In JAC, we deal with these scenarios by means of `Plasma.Computation`'s, a data type of be further explained and utilized below; cf. also Fig. 2.

A first plasma scenario, that makes heavily use of atomic input, is the modeling of ionic mixtures in a SB equilibrium, though for different plasma temperatures and matter densities. Other scenarios (might) refer to the estimation of line (and intensity) shifts in different empirical plasma models or environments, the reconstruction of atomic line shapes in plasma because of microscopic field distributions, or the computation of atomic partition functions, to make a few possible applications explicit. Most of these applications are indeed based on a given set of reference configurations, from which sizeable groups of atomic states, energies and properties need then to be generated automatically. Analogue to the cascade computations above, these scenarios can be assembled into several (plasma) schemes in order to make their implementation feasible. The goal of all these developments is to support selected plasma computations without much prior knowledge of the code.

While a few of these plasma schemes have already been implemented in JAC, others still need to be worked out. A complete realization of these plasma schemes is a task, much larger than what we can (and intend to) realize in the present version of JAC. The code can be readily forked from the web [36], including the present upgrade of the toolbox, and expanded towards other plasma applications (schemes). In our examples below, we shall explain the implemented *Saha-Boltzmann scheme* for calculating ionic mixtures in a SB equilibrium. It adds a SB solver to JAC in order to compute the chemical potentials and its population of excited states of a thin plasma by solving the Saha equation for the ionization equilibrium and the Boltzmann distribution for all excited levels. Care has been taken to keep this extension to JAC flexible, extensible and computationally reasonable efficient. Below, we explain how readily we can incorporate atomic data (ionization energies, energy levels, partition functions), to ensure charge neutrality or to handle multiple elements or provide output suitable for further analysis.

Table 2 Selected data structures of the JAC toolbox for plasma computations that are (all) based on a spherical cell. Here, only a brief explanation is given for each data type, whereas further details can be found from Julia's help facilities [49]

Struct and brief explanation
`Basics.AbstractPlasmaModel`: an abstract type to deal with different plasma models, such as `NoPlasmaModel`, `DebyeHueckel`, `IonSphere`, `StewartPyattModel` or `WithoutAutoionizationModel`
`Basics.IsotopicFraction`: a type to deal with different – sets of individual or averaged – isotopic fractions in an equilibrium computation
`Basics.Shell`: a enumeration for the allowed values of non-relativistic shell like $1s$, $2s$, $2p$,
`Basics.StewartPyattModel`: specifies – the parameters of – a Stewart-Pyatt (SP) plasma model in terms of the radius $r^{(SP)}$, the electron density $n_e(T)$ and the screening parameter $\lambda^{(SP)}$.
`Plasma.AbstractPlasmaScheme`: an abstract data type to distinguish different kinds of plasma computations; cf. Fig. 2 (middle panel)
`Plasma.AverageAtomScheme`: allows and supports to perform average-atom computations for dense plasma and based on a spherical cell
`Plasma.Computation`: a type for doing different (simple) plasma computation; cf. Fig. 2 (upper panel)
`Plasma.IonicClass`: a struct to comprise the (necessary) information about a single ionic charge state of a given isotopic fraction
`Plasma.IonicLevel`: a struct to comprise the (necessary) data about a single ionic level, including information about the uppermost shell in the underlying configuration
`Plasma.IsotopeClass`: a struct to comprise all (necessary) data about a single isotopic fraction
`Plasma.SahaBoltzmannScheme`: a struct to compute the chemical potentials and level population of ionic mixtures in SB equilibrium, or to determine their thermodynamic properties; cf. Fig. 2 (lower panel)
`Plasma.Settings`: specifies the plasma conditions, i.e. temperature, matter density and the use of number densities
`Radial.Grid`: defines the radial grid that contains all information about the grid parameters, the generation of the B-spline basis of all occupied radial orbitals as well as for performing radial integrations

3.3 Data structures useful for plasma simulations

In computational physics (and beyond) well-chosen data structures are known today as one of the corner stones for implementing efficient code and for enlarging its life cycle. These data structures typically form the building blocks to accurately express the physical requirements, whereas specific abstract data types help establish a hierarchy of relationships between data and actions by just providing a set of (concrete) subtype arguments. In the present expansion to JAC, we make carefully use of such relationships in the code in order to keep it expandable towards new schemes with-

out the need to modify the previously implemented ones. In total, there are at present about ~ 250 specifically designed abstract and concrete data structures in JAC, though most of them will remain hidden to the user of the toolbox.

In Julia, all types are indeed considered to be first-class and are used to dynamically select code at runtime. This is accomplished through multiple dispatch, which dynamically help select and execute the most specific method, suitable for the given types of arguments. Multiple dispatch also reduces the needs for argument checking at the begin of a function and, thus, allows programs to be better aligned with the underlying physics [48].

While we shall not lay out here technical details about the various plasma schemes, the module `Plasma` provides several data structures which are specific to the computation of plasma properties. Figure 2 displays the internal definition of three of these data structures that are directly accessible to the user. For instance, a `Plasma Computation` (upper panel) has been defined in order to deal with all information about the selected scheme, the grid as well as the basic settings (temperature, matter density) of a plasma. If, moreover, the computations just refer to the plasma properties of some isolated atom or ion, that is to be treated within a certain plasma environment, this type also designates the nuclear model and a list of reference configurations as applied for modeling the modified (atomic) level structure in a plasma. All further details of the plasma environment are then typically set out by means of the particular plasma scheme and its subfields.

The type `Plasma.AbstractPlasmaScheme` (middle panel of Fig. 2) helps distinguish the particular plasma scheme and facilitates the dynamic assignment of the code as associated with any of its concrete subtypes. While a few concrete types are briefly indicated in this panel, we shall consider here only the `Plasma.SahaBoltzmannScheme` (lower panel of Fig. 2) that enables one to solve the SB equilibrium self-consistently for different ionic mixtures and (non-ideal) plasma conditions. The underlying isotopic mixture $\{(Z, A, x)\}$ can be provided as a list of (non-normalized) isotopic fractions, whereas the relevant charge states $q+$ of the ions are either left unspecified (and are determined later during the self-consistent iteration) or can be restricted also by a `qRange::UnitRange{Int64}` in order to limit the computations. This choice makes this scheme well suitable to deal with mixtures of medium and heavy elements. Beside of the automatic generation of all atomic input of interest, these (level) data can be provided also by a set of data files (for instance, from some previous run of the code) for selected or all isotopes (Z, A) in the mixture. Other subfields of the `Plasma.SahaBoltzmannScheme` will be explained in the next section below. For each data structure in JAC, a detailed description of these data types can usually be obtained interactively, for instance by `? Plasma.Computation`, in order to quickly recall the purpose of this `struct` along with the definition of all associated subfields.

To better understand the example below, we will need a few further data types as compiled in Table 2. Two prominent types from this table are an `IonicLevel` to comprise the information about a single level, including a few details about the underlying computational approximation, as well as an `IonicClass` to keep the data about a single ionic charge state together. Such hierarchies of data structures are frequently used in JAC to design types in a form that clearly reflect the "building blocks" of the program and, hence, help reduce much of the technical slang (which is otherwise quite common to most codes in atomic and plasma physics). As seen from table 2, we aim to apply notions that are – more or less – readily understandable to most atomic and plasma physicists as well as to potential users of the code.

3.4 Running Saha-Boltzmann plasma computations for ionic mixtures

Any SB plasma computation generally aims first of all to determine at equilibrium the level populations and charge-state distribution of all isotopic fractions in the mixture. Apart from implementing a reasonable efficient (numerical) scheme for solving the equations, this population depend of course on the underlying models that are employed (i) for representing the (low-lying) atomic level structure of all ions and (ii) the requested corrections for dealing with a non-ideal plasma. These computational models account for the energy functional used in the self-consistent-field of the ions, the truncation of levels in the evaluation of partition functions, limitations of the atomic Hamiltonian or the detailed coupling of the electron continuum due to contributions from the surrounding plasma, to mention a few. To characterize these (computational) models and the amount of ionic level data in JAC, we provide simple access to the following parameters: (1) The number of electronic excitations (`NoExcitations`; i.e. single, double,... excitations) to be included with regard to an empirically determined ground (reference) configuration of the ion $[I(Z, A)]^{q+}$; (2) the maximum principal n-shell quantum number (`upperShellNo`) used in the representation of the ionic levels; (3) the range of charge states of the ions in the mixture and (4) the maximum number of ionic levels (`maxNoLevels`) that are included for any ion $[I(Z, A)]^{q+}$. All these numbers are kept also in the ionic-level data files in order to ensure consistent equilibrium simulations.

A SB (plasma) scheme also enables one to specify the plasma model and, hence, the way how IPD is taken into account internally. Several models are available in JAC for dealing with IPD, including the `DebyeHueckelModel()`,

`IonSphereModel()` and the `StewartPyattModel()`. We also allow to consider ion mixtures without any IPD, the `NoPlasmaModel()`, or by just excluding all autoionizing levels in the computation of the partition functions (`WithoutAutoionizationModel()`). The physical assumptions and conditions of these plasma models were summarized in Table 1 above. Since these models are all subtypes of `Basics.AbstractPlasmaModel`, this list could be readily extended if such needs arise from the user's side.

4 Saha-Boltzmann solutions for selected ionic mixtures. Simple examples

The present expansion of the Jac toolbox has been designed flexibly enough to deal with ionic mixtures of (almost) all elements across the periodic table. In particular, this supplement helps to analyze the equilibrium populations and charge-state distributions for mixtures from a wide range of astrophysical objects. Although, for the sake of simplicity, we shall restrict our examples below to rather simple and partly even single-isotopic mixtures of ions, this flexibility in selecting such mixtures will facilitate more complex simulations in the future. Further work on more complex ionic mixtures will be needed to better understand the range in which our approach is valid and how it compares with other computational tools.

4.1 A high-temperature carbon plasma

Carbon has been found essential for the formation and diversity of different astrophysical objects. In the stellar evolution, for instance, carbon is known to be formed via the (so-called) triple-alpha process in course of the helium-burning phase [50]. For massive stars, this burning then initiates also the nucleosynthesis of heavier elements, such as neon, magnesium, sodium and, hence, the chemical enrichment of most of the universe. From the carbon-to-oxygen ratio, moreover, one may predict the further evolution of stars in terms of Type Ia supernova, neutron star or black hole [51]. In addition, the absorption lines of C^+ and C^{2+} ions have been frequently employed to extract the gas temperature, density, and ionization states in different astrophysical environments [52].

With this broad interest in mind, let us start with a (pure) carbon plasma at high temperature $T \approx 100$ eV. At this temperature, most carbon atoms will be doubly or more ionized owing to electron-impact, photoionization and capture processes, and as readily estimated from its second ionization potential $I_p^{(2)} \approx 24.4$ eV. For a weakly-coupled plasma, moreover, the ionization potentials of free ions will be lowered and, thus, may likely enhance the mean charge state of the plasma. For a mixture of carbon ions, for example, the overall IPD contributions can be readily analyzed by running Jac with the input example from Fig. 3. Apart from the radial grid, this input first designates by means of the `Plasma.Settings` the matter density ρ [g/cm^3], plasma temperature [Hartree] as well as the logical flag (true) in order to indicate that $\rho \equiv n_i$ [ions/a_o^3] represents instead the ion number density throughout the simulations.

As outlined above, the `ionMixture` in this (first) input just comprises a single C ($Z = 6.$, $A = 12.2$) isotope, though further isotopes could be added freely. In the `SahaBoltzmannScheme`, we later combine all these parameters together with the information about the – generation and/or use of the – ionic level data. For carbon ions, these data are provided here by the given filename and for the charge states $q = 0 .. 6$. In addition, we have limited the maximum number of levels for each charge state to 10.000 due to the account of single and double excitations (with regard to the ground configuration of each ion) within the wave function expansions of all considered ions. Finally, moreover, we allow for the occupation of all shells with principal quantum number $n \leq 8$ in these wave functions.

All these physical input parameters are finally merged into the (specification of the) `Plasma.Computation`. In this assignment, we make use of a frequently implemented constructor for (complex) data types in Jac by just overwriting those parameters that differ from their default values, and as given by the plain constructor `Plasma.Computation()`. Once, such a (plasma) computation has been has been assigned to some variable `comp`, all what is left to the user is to `perform(comp::Plasma.Computation)`. Obviously, each such plasma computation will lead to a – more or less – sizeable set of output data which, along with some standard printout to screen, can be utilized in further steps of the plasma modelling. Moreover, if additional ionic level data are generated by the code in course of a given computation, the produced data files will be stored automatically to disk as well. No further control parameters occur at present in the input to a SB equilibrium computation but could be easily incorporated (along with the necessary code to be implemented in this case) by providing, for instance, extra subfields to the `scheme::Plasma.SahaBoltzmannScheme`.

In the input above, we can account for the IPD of a plasma by choosing a proper (plasma) model in the set-up of the `scheme`. For a Stewart-Pyatt plasma model, that is suitable for dense and high-temperature environments, we could use `StewartPyattModel(ne,lambda)` to enter the electron density and the Debye length for this model. In practice, however, these parameters are usually determined only in course of the computations. The IPD becomes significant especially if the mean free path of electrons is short and the ions are packed rather tightly to each other like in white dwarfs, stellar envelopes, or the interiors of giant planets. The simple handling of IPD in Jac ensures that the SB computa-

```
# Compute the Saha-Boltzmann equilibrium densities for a mixture of Carbon ions
grid        = Radial.Grid(Radial.Grid(false), rnt = 2.0e-5, h = 5.0e-2, hp = 0.8e-2, rbox = 25.0)
nm          = Nuclear.Model(6.0)
nIon        = 1.0e-3                # ion number density
temp_au     = Defaults.convertUnits("energy: from eV to atomic", 100.)
settings    = Plasma.Settings(temp_au, nIon, true)
ionMixture  = [IsotopicFraction(6., 12.011, 1.0)]
scheme      = Plasma.SahaBoltzmannScheme(Basics.NoPlasmaModel(), true, true, 0:6, 10000, 2, 8,
                                         ionMixture, String["IonicLevelDataZ6A12.jld"])

comp        = Plasma.Computation(Plasma.Computation(), scheme=scheme, grid=grid, settings=settings)
results     = perform(comp, output=true)
```

Fig. 3 Input to JAC for running a `Plasma.Computation` that calculate the SB equilibrium of some mixture of (only) carbon ions with number density $\rho \equiv n_i = 1 \times 10^{-3}$ ions/a_o^3 at plasma temperature $T = 100$ eV. The `SahaBoltzmannScheme` rules the code and provides information about the plasma model, ionic mixture, range of charge states as well as several other parameters to readily control the execution. See text for further explanations

tional model reflects the real plasma environments as found in stellar interiors, fusion experiments or elsewhere.

Practically, of course, the charge state (distribution) and thermodynamic properties at SB equilibrium will depend also on the energies and explicit representation of all ionic levels involved. In JAC, in fact, the major (computational) effort typically refers to the calculation of the fine-structure levels as well as the numerical determination of level energies. The generation of these levels can be further optimized by the user, if the default `ManyElectron.AsfSettings()` is overwritten for the given `Plasma.Computation`. However, while the JAC code enables one to select many of these physical parameters (more or less readily), this control requires some insight and need usually to be further explored by additional computations. For many applications, therefore, the defaults are likely a good starting point but should be augmented by additional computations and perhaps the comparison with other tools in order to benchmark the results and to understand their sensitivity with regard to selected model parameters. Whereas the individual (low-lying) level energies are meanwhile well understood for most atoms and ions, this does not necessarily apply for the computation of partition functions, where thousands of such (excited) levels are required simultaneously.

The upper panel of Fig. 4 displays the relative charge-state population of a (pure) high-temperature carbon plasma at ion number density $n_i = 10^{-3}$ ions/a_o^3. Results are shown for the two temperatures $T = 50$ eV (red diamonds) and $T = 100$ eV (blue triangles), respectively- For $T = 100$ eV, moreover, the contributions from IPD in a Stewart-Pyatt plasma model are shown (cyan triangles) together with analogue data from an average-atom model computation ([53]; green circles). The lines between the symbols at well-defined charge states in Fig. 4 are just displayed to guide the eyes but without having any further meaning.

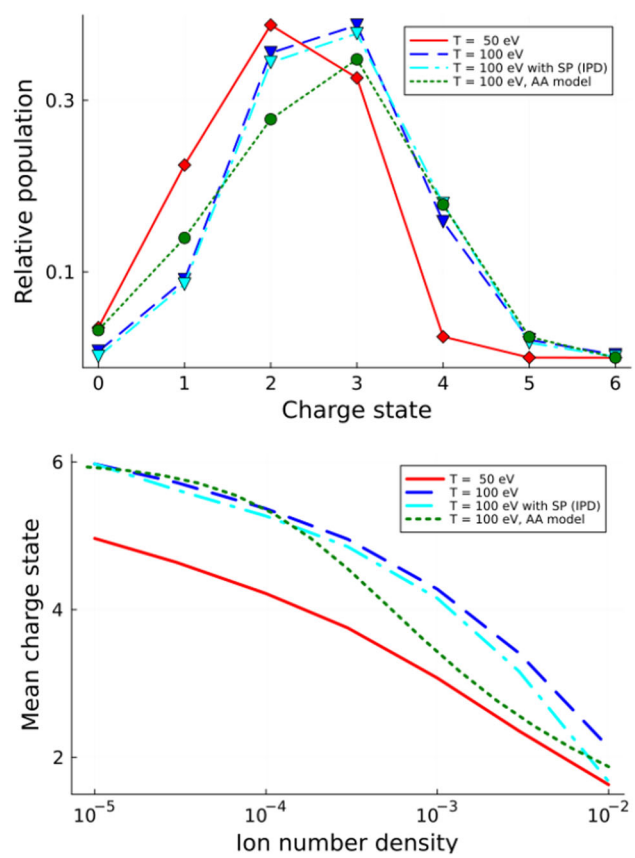

Fig. 4 Charge-state distribution of a (pure) high-temperature carbon plasma. Upper panel: relative population of charge states for a plasma at ion number density $n_i = 10^{-3}$ ions/a_o^3 and at temperatures $T = 50$ eV (red diamonds) as well as $T = 100$ eV (blue triangles). For $T = 100$ eV, moreover, the contribution from IPD in a Stewart-Pyatt plasma model are shown (cyan triangles) together with analogue data from an average-atom model computation ([53]; green circles). The lines are shown here only to guide the eyes. Lower panel: Mean charge state as function of the ion number density n_i [ions/a_o^3] and for the same temperatures as well as colors/line styles as in the upper panel above. See text for further discussion

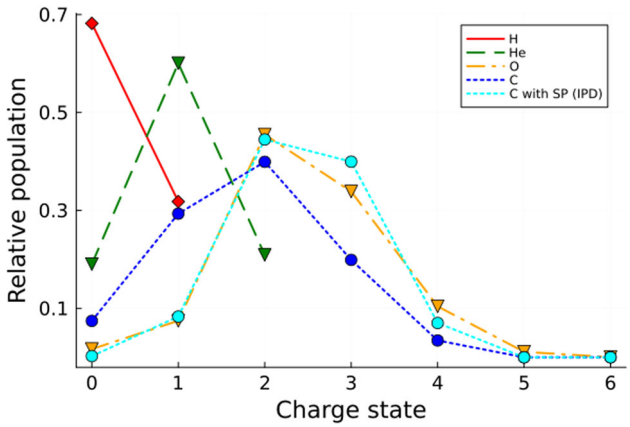

Fig. 5 Charge-state distribution of a (simplified) ionic mixture from the solar photosphere with abundances $A_{\mathrm{H}} = 0.73507$, $A_{\mathrm{He}} = 0.2470$, $A_{\mathrm{O}} = 9.2230 \cdot 10^{-3}$, $A_{\mathrm{C}} = 4.4144 \cdot 10^{-3}$ at temperature $T = 2.77 \cdot 10^5$ K and matter density $\rho_m = 8.136 \cdot 10^{-3}$ g/cm^3 [54,55]. The relative population of charge states are shown separately for all selected elements H (red diamonds), He (green triangles), O (orange crosses) and C (blue circles). For carbon, moreover, the shift due to the IPD in a Stewart-Pyatt potential is shown for C (cyan circles). The lines are shown again to guide the eyes

As expected, the charge state distribution are shifted towards higher charge states as the temperature increases.

A similar, though much smaller, shift is seen also for a Stewart-Pyatt potential due to the lowering of the ionization potential, simply since more electrons will belong to the continuum. While, for a low-density plasma, the effect of IPD remains moderate, the shift becomes larger if both, the temperature and/or ion number densities increases. All these data have been generated by using repeatedly input very similar to Fig. 3. Analogue charge-state distributions can be readily computed for other temperatures, ion densities or if the IPD model is changes, including also simply the modified treatment of the autoionizing levels in the computation of partition functions.

In the lower panel of Fig. 4, we exhibit the mean charge state of the carbon plasma as function of the ion number density n_i at logrithmic scale. Lines (with the same colors and style as in the upper panel) are shown again for the temperatures $T = 50$ and 100 eV. A strong decrease of the charge state is seen and fulfills our forecast about a closely-packed plasma. This lower panel also compares our solutions of a SB equilibrium with the IPD from a Stewart-Pyatt plasma model and with the average-atom computation by Piron [53]. First steps have been undertaken to incorporate some comparable average-atom model directly into Jac.

Several uncertainties in choosing the parameters of the underlying computational model occur naturally and will affect the results in Fig. 4 and in most analoge calculations. These uncertainties arise from (a) the mere number and the representation of the atomic levels involved, (b) the range of selected charge states of the ionic mixture, (c) the choice of the plasma model for given temperature and matter density as well as (d) from yet omitted plasma contributions, such as the treatment of particle correlations or omitted recombination processes. We have implemented this expansion to the Jac code for supporting the analysis of these uncertainties without being able by now to make them explicit over a wide range of plasma conditions.

4.2 Ion mixtures from the solar photosphere

Much attention has been paid over the years to the chemical composition of the sun to better understand the formation and evolution of the Solar System. This solar photospheric composition has been widely used as a standard for studying the chemical evolution of the Milky Way. The elemental abundances in the photosphere help estimate the primordial material from which the Sun and solar system was formed originally. The chemical composition of the solar photosphere has undergone slight but measurable changes over time owing to the atomic and nucleo-synthesis processes that keep the sun "burning". Whereas these changes are rather moderate, when compared to its initial composition about 4.6 billion years ago, they have important implications for astrophysics.

While the photosphere of the sun primarily consists out of hydrogen and helium, oxygen, carbon and other light metallic ions play a critical hole for the opacity and energy transport in different layers of the sun. A lower metallicity of some layer typically reduces the opacity and therefore directly affects the internal structure. For the solar photosphere, the abundances of light metallic elements have changed recently [54] and have lead to a modified ratio of carbon and oxygen, relative to hydrogen [55,56]. Many of the applied solar models are hereby based on the OPAL opacity tables [57] and the detailed knowledge of the (mean) charge-state distributions in the solar photosphere.

To explore and better understand these charge distributions in the photosphere of the Sun, we can apply Jac's SB solver by using the abundances of Bahcall and coworkers [54]. For simplicity reasons, we here consider an ionic mixture with the abundances [55] $A_{\mathrm{H}} = 0.73507$, $A_{\mathrm{He}} = 0.2470$, $A_{\mathrm{O}} = 9.2230 \cdot 10^{-3}$ and $A_{\mathrm{C}} = 4.4144 \cdot 10^{-3}$, respectively. For such a (fixed) mixture, Fig. 5 displays the relative population of charge states separately for all selected elements H (red diamonds), He (green triangles), O (orange crosses) and C (blue circles). For carbon, moreover, the shift due to the IPD in a Stewart-Pyatt plasma model is shown also explicitly. Whereas the charge-state distribution peak at $q = 2$ for both metallic ions (O and C), carbon exhibits a higher population of the $q = 1$ and the neutral charge state, when compared to oxygen. However, these distributions are centered around $q = 3\,(4)$ if the IPD is taken into account into the modeling. Further studies on (metallic) ionic mixtures are planned to resolve more details about the choice

of plasma models and how much of the shifts are already obtained, if the autoionizing levels are consequently omitted in the computation of the partition functions.

4.3 Towards mixtures of heavy metallic ions

Despite of their low abundances, metallic ions play a key role in the evolution of the Universe. In many astrophysical objects, these ions severely affect the ionization state of plasma because of their contribution to a variety of collisional as well as photo-ionization and recombination processes. Moreover, the absorption (and scattering) of light by metallic ions also regulates how energy is released from stellar atmospheres.

In the core of massive stars, for instance, multiply-charged iron ions often dominate the opacity and energy transport. Recently, medium and heavy metallic ions have been found prominent also for understanding kilonovae and neutron-star mergers, where they are produced by a rapid neutron capture and the beta decay of successively heavier isotopes. Especially the lanthanides with atomic numbers $Z = 57..71$ seem to slow down the radiation transport owing to their complex fine-structure and, hence, to lead to longer-lasting and redder light curves.

Therefore, different plasma simulations are highly desirable to address various extreme plasma conditions that (may) arise during the formation and evolution of heavy metallic ions. For example, emphasis should first be placed upon the EOS of metallic ions including plasma at high temperatures and densities, for which IPD and the degeneracy of the electron states are supposed to be relevant. These simulations will help comprehend the energy deposition and thermalization of (radioactive) decay products and how metallic ions affect the spectral output of the merger events.

5 Conclusions and outlook

The JAC toolbox has been expanded in this work to simulate the (mean) charge-state composition and thermodynamic properties of ionic mixtures under certain plasma conditions. Emphasis has been placed to establish a relativistic SB solver, suitable for medium and heavy elements as well as on models for dealing with IPD and further corrections. When compared with other SB solvers, the present expansion of the JAC toolbox is based on various (carefully selected kinds of) `Plasma.Computation`'s and, hence, the direct use of *ab-initio* atomic computation as central ingredient in the plasma simulations. It also supports a simple control of ionic mixtures in different approximations and by incorporating useful corrections due to microscopic interaction among the plasma particles.

With the present expansion of the code, the JAC toolbox will help address several key processes and phenomena in different astrophysical environments. A proper knowledge of the ionic level distribution and partition functions is known to be crucial for interpreting the emission and absorption lines in the astrophysical spectra. In the future, the present SB equilibrium computations with IPD might be combined also with cascade computations and collisional-radiative (CR) models to account for non-LTE contributions, either in a static CR equilibrium or dynamically as time-evolution of populations. The estimation of the underlying rates and cross-sections of radiative processes, such as photo-excitation and de-excitation, photoionization as well as recombination or electron-impact processes is at the heart of JAC and will help to advance the plasma modeling in the future [58,59].

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Data Availability Statement All data in Figs. 4 and 5 can be reconstructed by means of the JAC toolbox, provided at the website [36].

Code Availability Statement This manuscript has no associated code/software but this is provided at the website [36].

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